Comprehensive Technical Literature Review & Analytical Study (10 Core Papers) — Neural Mechanical Digital Twin (NMDT) Knee Exoskeleton

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Prepared by: Eng. Mohamed Zedan | Full-Text Verified Mathematical Formulations, Sensor Architectures, Experimental Protocols & Journal Quartiles

Executive Master Matrix: 10-Paper Comparison (Verified Quartiles, Payloads, Active vs. Monitoring Sensors & Results)

#	Paper & Year	Journal	Verified Quartile	Payload & Subjects	Active Control Sensors (In-Loop)	Offline Monitoring / Validation Sensors	Key Quantitative Results
1	Arefeen et al. (2024)	Biomimetics	**Q2 (SJR 0.66) / Q1 (JCR)**	10 kg box (Sim + N=1 Exp)	**Knee Encoder**	Xsens MoCap, 2 Force Plates, 6-ch sEMG	Sim: -11.9% knee, -38% shoulder torque; Exp: -16.8% to -30.1% quad sEMG, -35% back sEMG
2	Divekar et al. (2024)	Science Robotics	**Q1 (SJR 7.15 / Top 1%)**	Weighted Box LLC (N=10)	**4 IMUs, Motor Encoders, Insole FSRs**	5-ch sEMG (VMO, VL, RF, BF, ST)	-14.5% overall quad sEMG (-28% lifting); -43% post-fatigue lift time deficit
3	Meagher et al. (2025)	IEEE Access	**Q1 (SJR 0.96) / Q2 (JCR)**	9 kg box squat lift (N=8)	**Absolute Knee Encoder (MU18S)**	19 Vicon Cams, 3 Force Plates, VM/VL sEMG	-53.4% / -53.58% peak biological knee torque; -29.2% to -31.2% peak VM/VL sEMG
4	Yu et al. (2025)	Ann. Biomed. Eng.	**Q2 (SJR 0.73 / JCR Q2)**	Bodyweight squats (N=7)	**Thigh/Shank IMUs + Motor Encoders**	Cosmed Mask, Polar H10 HR, 5-ch sEMG, RPE	At β=50%: -12.8% metabolic, -13.8% HR, -39.4% to -43.2% quad sEMG (β=100% increased BF +26.6%)
5	Chang et al. (2026)	Front. Bioeng. Biotech.	**Q1 (SJR 1.15 / JCR Q1)**	14, 28, 42 Nm vs Barbell (N=15)	**None (Pre-set Mechanical Clutch)**	3D MoCap, 2 Force Plates, 5-ch sEMG, OpenSim	+20% to +40% hip power shift, -20% to -30% knee power, -50% VM sEMG, but -7° to -9° ROM loss
6	Dai et al. (2025)	IEEE TMRB	**Q1 (SJR 0.85) / Q2 (JCR)**	60° Squat Hold 30s, 0 kg (N=7)	**2 IMUs, 18-bit Encoder, 6-Axis Force Sensor**	16-marker Vicon, 2 Force Plates, 8-ch sEMG, CEINMS	-14.53% knee moment, -29.00% joint contact force, -24.40% quad muscle force, -23.48% sEMG
7	Boon (2025) / Zhang (2021)	IEEE RA-L / Sci. China	**Q1 (SJR 1.56 / JCR Q1)**	9.1 kg Vest (N=8) / 30 kg Load	**Encoder+IMU (Boon) / +2 Force+Insoles (Zhang)**	8-ch sEMG (Boon) / HR & Lungs (Zhang)	Boon: -32% peak & -29% mean quad sEMG | Zhang: 80% knee torque compensated, <0.2s Lyapunov switch
8	Xia et al. (2024)	Sensors	**Q1 (SJR 0.81) / Q2 (JCR)**	12-Step Stair Ascent (N=5)	**6 Wearable IMUs (200 Hz) + 2 Torque Sensors**	8 Vicon Cams + Kistler Plates (Train), RF/VM sEMG	RMSE 4.64e-3 Nm/kg at t_s=50ms; RF -20.87%, VM -17.45% (-19.37% avg quad sEMG)
9	Zhao et al. (2025)	IEEE/ASME TMECH	**Q1 (SJR 2.54 / JCR Q1)**	Actuator Bench & Tracking	**Dual Optical Encoders + SEA Spring Sensor**	None (Electromechanical Tracking Only)	0.31 Nm RMS torque tracking error; 1.03 Nm transparent-mode resistive torque
10	Monteiro et al. (2024)	Sensors	**Q1 (SJR 0.81) / Q2 (JCR)**	Treadmill 1.5–3.0 km/h (N=5 / N=1)	**4 IMUs (Chest/Wrist/Waist/Ankle) + Strain Gauges**	Cosmed K5 Mask (Protocol 1 Validation Only)	10s EGPR RMSE 0.66 W/kg; HITL CMA-ES: -7.3% metabolic cost, -32.3% interaction torque

💡 Executive Architectural Synthesis: How the 10 Reviewed Studies Justify Our Proposed NMDT Framework

1. Hardware Selection — Why QDD Actuation (Papers #2, #4, #8)?
Low-ratio (9:1–10:1) QDD brushless actuators offer ultra-low backdrive resistance (0.5–2 Nm) and high bandwidth, avoiding the 1.6 km/h speed ceiling of 160:1 Harmonic Drives (Paper #10) and the 7°–9° ROM loss of uncompensated mechanical impedance (Paper #5).

2. NMDT Module A (Real-Time Musculoskeletal Digital Twin):
Existing studies either solve coupled human-motor dynamics offline (Paper #1), assume static body-weight springs (Paper #3), or run EMG-informed CEINMS models offline (Paper #6). Module A runs a low-latency biomechanical digital twin online.

3. NMDT Module B (Hybrid CNN-LSTM + 50 ms Time-Shifted Reference):
Upgrading Xia et al. (Paper #8), Module B predicts biological knee torque 50 ms ahead of actuator phase lag while enforcing Yu et al.'s (Paper #4) asymmetric eccentric-to-concentric ratio (β = 50%) so hamstrings never fight the exoskeleton during squat descent.

4. NMDT Module C (Active sEMG Fatigue-Adaptive Modulator):
Across Papers #1–#8, sEMG was used strictly as an OFFLINE monitoring metric—never inside the active feedback loop! And HITL CMA-ES (Paper #10) requires 20 minutes to converge. Module C closes the loop with live sEMG fatigue indices (RMS/MDF) on a single-lift timescale.

PAPER #1 (P1)

Q2 (SJR: 0.664) / Q1 (JCR IF: 4.2)

Human–Exoskeleton Coupling Simulation for Lifting Tasks with Shoulder, Spine, and Knee-Joint Powered Exoskeletons

Authors: Asif Arefeen, Ting Xia, and Yujiang Xiang (Oklahoma State University & Northern Illinois University, USA) | **Year:** 2024 | **Journal:** Biomimetics (MDPI), Vol. 9, Issue 8, Art. 454 | **DOI:** 10.3390/biomimetics9080454

SCImago (SJR): Q2 in Bioengineering & Biomaterials (SJR = 0.664) | Clarivate Web of Science (JCR): Q1 in Multidisciplinary Sciences (IF = 4.2)

1. Core Idea & Executive Overview

- Problem Statement: Repetitive manual box lifting generates excessive lumbosacral (L4–L5) spinal disc compression and lower-limb muscular fatigue, representing a primary cause of occupational musculoskeletal disorders.
- Proposed Solution: A predictive 2D sagittal musculoskeletal–exoskeleton dynamic simulation (10-DOF human model coupled with electromechanical DC motor dynamics) that optimizes lifting trajectories and assistive joint torques. Furthermore, Sections 4.1 & 4.2 present an experimental validation on a human subject wearing a commercial powered knee exoskeleton (Picasso Intelligence, LLC) while lifting a 10 kg box.

2. Primary Research Question (RQ)

How can a coupled human–exoskeleton dynamic model (integrating Denavit–Hartenberg skeletal kinematics and recursive Lagrangian dynamics with DC motor electromechanical equations) synthesize optimal lifting trajectories that minimize sum-of-squared human joint torques, and does applying the optimized knee torque profile to a physical knee exoskeleton reduce quadriceps and lumbar erector spinae sEMG activation?

3. Hardware, Mechanical Design & Actuation Specifications

- Simulation Model: 2D sagittal 10-DOF human skeletal model formulated via Denavit–Hartenberg (DH) parameters coupled with armature-controlled DC motors and gearboxes at the shoulder, spine/hip, and knee joints.
- Experimental Hardware (Sec. 4.1): Powered knee exoskeleton manufactured by Picasso Intelligence, LLC — Mass: 2.25 kg; Peak Torque Limit: [-16, +16] Nm; actuated by a DC/BLDC motor with an integrated optical joint angle encoder.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: Integrated knee joint angle optical encoder (θ_knee).
- Laboratory Validation Suite: Xsens 3D inertial motion capture system (60 Hz), two Bertec force plates (1000 Hz), and 6-channel Delsys Trigno wireless sEMG (2000 Hz, band-pass 20–450 Hz) on right Rectus Femoris (RF), Vastus Lateralis (VL), Vastus Medialis (VM), Biceps Femoris (BF), Erector Spinae (ES), and Latissimus Dorsi (LD).

B) Computed / Predicted Outputs

- Simulation Optimization Outputs: Optimal joint trajectories q(t), ground reaction forces (GRF), biological human joint torques τ_h(t), motor armature currents/voltages (I, V), and optimal assistive torque profile τ_e(t).
- Hardware Controller Output: Real-time target assistive knee torque τ_e(θ_knee) generated via cubic B-spline interpolation of live encoder angle (up to 16 Nm peak).

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: Only the integrated knee joint angle encoder (θ_knee) drives the closed-loop B-spline assistive torque law τ_e(θ_knee).
- OFFLINE MONITORING & VALIDATION: Xsens MoCap, Bertec force plates, and 6-channel Delsys sEMG are used strictly offline to validate kinematic trajectories and quantify muscular effort reduction.

5. Control Architecture & Exact Mathematical Equations

- Eq. (1) — Coupled Human–Exoskeleton Recursive Lagrangian Dynamics (DH Formulation):

$$\tau_{h,i} = \mathrm{tr}((\partial A_i/\partial q_i)\, D_i\, A_i^T) - g^T (\partial A_i/\partial q_i)\, E_i - f_k^T (\partial A_i/\partial q_i)\, F_i - G_i^T A_{i-1} Z_0 - \tau_{e,i}$$

- Eq. (2)–(6) — DC Motor Electromechanical Dynamics:

$L(dI/dt) = V - K\cdot\dot{\theta} - R\cdot I$ (Eq. 2) | $T_{motor} = K\cdot I$ (Eq. 3) | $T_l = T_{motor} - J_m\cdot\ddot{\theta} - b\cdot\dot{\theta}$ (Eq. 5) | $\tau_e = GB_r \cdot T_l$ (Eq. 6)

- Eq. (7) — Nonlinear Optimal Control Objective Function (SNOPT SQP solver with n_ctrl = 5 B-spline control points/joint):

$$\min J = \int_0^T \sum_{i=1}^{n_{dof}} (\tau_{h,i} / \tau_{h,i^{max}})^2\, dt$$ (Minimizing sum of normalized squared human joint torques)

- Hardware Control Mapping: Least-squares cubic B-spline interpolation $\tau_e = f(\theta_{knee})$ driven by live encoder feedback.

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Simulation Cases: Symmetric squat lifting of a 10 kg box (0.31 × 0.31 × 0.42 m) from 0.1 m initial height to 1.0 m waist height over T = 1.6 s across 5 configurations: No Exo, Knee Exo (±16 Nm), Hip/Spine Exo (±30 Nm), Shoulder Exo (±30 Nm), and Combined 3-Joint Exo.
- Experimental Hardware Trial: N = 1 healthy male participant (Age: 27 yrs, Height: 1.70 m, Mass: 68.75 kg) lifting a 10 kg box for 3 repetitions with vs. without the Picasso knee exoskeleton (16 Nm peak torque).

7. Main Quantitative Results & Statistical Significance

- Simulation Peak Biological Torque Reductions (Combined 3-Joint Exo vs. No Exo): Shoulder -38.01%, Lumbar Spine (L4–L5) -6.40%, Knee -11.93% (SNOPT convergence time: 1.31–13.79 s).
- Experimental 10 kg Squat-Lift Peak sEMG Reductions (Knee Exo vs. No Exo, N = 1):
  - Lumbar Erector Spinae (ES): -35.00%
  - Vastus Lateralis (VL): -30.14% | Biceps Femoris (BF): -25.71%
  - Vastus Medialis (VM): -22.12% | Rectus Femoris (RF): -16.77% | Latissimus Dorsi (LD): -10.03%

8. Internal Study Limitations

1. Planar 2D sagittal assumption neglects 3D coronal/transverse trunk torsion and asymmetric lifting loads.
2. Trajectory optimization is computed offline (up to 13.79 s CPU time) for a known static 10 kg box mass.
3. Experimental hardware validation was limited to N = 1 subject using a fixed angle-dependent lookup curve $\tau_e(\theta_{knee})$ without adaptive payload estimation or real-time muscle fatigue compensation.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Offline trajectory optimization cannot adapt when the lifted payload mass changes across tasks or as the worker's muscles fatigue. The proposed NMDT architecture addresses this gap by integrating an online Neural Torque Predictor (Module B) with a Real-Time EMG-Driven Musculoskeletal Digital Twin (Modules A & C) that continuously scales assistance with fatigue.

10. Key Technical Discussions & Methodological Clarifications

Q1: Scope of Validation — Simulation vs. Physical Hardware:

While the core formulation is a coupled 2D dynamic optimization, Sections 4.1 and 4.2 provide experimental validation on a physical participant (27 yrs, 68.75 kg) wearing a 2.25 kg Picasso Intelligence knee exoskeleton (16 Nm peak) lifting a 10 kg box across 3 repetitions, demonstrating 16.8%–30.1% quadriceps sEMG reduction and 35.0% lumbar erector spinae reduction.

Q2: Transfer of Offline Time-Indexed Optimal Torque to Real-Time Hardware:

In the SNOPT simulation, the assistive torque $\tau_e(t)$ is parameterized by time via cubic B-splines. To deploy this profile on physical hardware without time-synchronization drift, the authors fitted the simulated optimal knee torque as a cubic B-spline function of the knee flexion angle $\tau_e(\theta_{knee})$, enabling real-time closed-loop execution using only the onboard knee encoder.

Q3: Rationale for Embedding DC Motor Equations (Eq. 2–6) Inside Biomechanical Optimization:

Standard biomechanical simulations treat exoskeleton assistance as an idealized, massless torque source. Coupling the armature circuit $L(dI/dt) = V - K\dot{\theta} - RI$ and rotor inertia J_m directly into the nonlinear programming constraints guarantees that the synthesized assistive torque trajectory remains within physical motor voltage, current, and bandwidth limits.

PAPER #2 (P2)

Q1 (SJR: 7.150 / JCR Q1 Top 1%)

A Versatile Knee Exoskeleton Mitigates Quadriceps Fatigue in Lifting, Lowering, and Carrying Tasks

Authors: Nikhil V. Divekar, Gray C. Thomas, Avani R. Yerva, Hannah B. Frame, and Robert D. Gregg (University of Michigan & Texas A&M University, USA) | **Year:** 2024 | **Journal:** Science Robotics, Vol. 9, Issue 94, eadr8282 | **DOI:** 10.1126/scirobotics.adr8282

SCImago (SJR): Q1 #1 in Robotics (SJR = 7.150) | Clarivate Web of Science (JCR): Q1 in Robotics (IF ~ 26.1)

1. Core Idea & Executive Overview

- Problem Statement: Existing industrial exoskeletons are typically restricted to single, isolated tasks (assisting either squat lifting, level walking, or stair climbing alone) and lack the versatility to transition smoothly across compound warehouse regimens.
- Proposed Solution: The bilateral M-BLUE Quasi-Direct Drive (QDD) knee exoskeleton governed by a unified, task-adaptive phase-based controller that modulates virtual springs, dampers, and gravity/inertia compensation across compound Logistics Lifting-Lowering-Carrying (LLC) circuits under both non-fatigued and quadriceps-fatigued conditions.

2. Primary Research Question (RQ)

Can a backdrivable QDD knee exoskeleton governed by a continuous kinematic pseudophase variable mitigate quadriceps fatigue, reduce post-fatigue lifting completion-time deficits, and preserve postural stability across multi-modal lifting, lowering, level/inclined carrying, and stair navigation?

3. Hardware, Mechanical Design & Actuation Specifications

- Exoskeleton Platform: M-BLUE (Michigan Backdrivable Lower-Limb Unloading Exoskeleton), bilateral wearable design.
- Actuators: T-Motor AK80-9 Quasi-Direct Drive (QDD) brushless motors with a low 9:1 planetary gear ratio (no belts/cables).
- Torque & Backdrivability: 25 Nm peak assistive torque; ultra-low backdrive torque (< 2 Nm) allowing unimpeded free leg swing.
- Embedded Processing & Power: Raspberry Pi 4B high-level processor communicating with Dephy brushless motor drivers; 24 V battery pack.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: 4× 6-axis IMUs (mounted bilaterally on thighs and shanks measuring global segment angles θ_th, θ_sh) + 2× motor optical encoders (knee angle/velocity) + custom Force-Sensitive Resistor (FSR) shoe insoles (measuring GRF).
- Laboratory Evaluation Suite: 5-channel wireless sEMG on Vastus Medialis Oblique (VMO), Vastus Lateralis (VL), Rectus Femoris (RF), Biceps Femoris (BF), and Semitendinosus (ST), plus optical motion capture and task timing metrics.

B) Computed / Predicted Outputs

- Active Controller Outputs: Continuous kinematic pseudophase variable φ, insole stance load factor, and target torque command converted directly to motor current (open-loop current control up to 25 Nm peak without a series torque sensor).
- Evaluation Outputs: Quadriceps sEMG amplitude (%MVC), Median Frequency (MDF) fatigue shifts, and lift completion time.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: 4× Thigh/Shank IMUs + Motor Encoders + Insole FSRs (drive the Energy Shaping & Pseudophase law).
- OFFLINE MONITORING & EVALUATION: 5-channel sEMG (VMO, VL, RF, BF, ST) and MDF fatigue metrics are used strictly offline to quantify fatigue mitigation; sEMG is not used inside the active feedback controller.

5. Control Architecture & Exact Mathematical Equations

- Continuous Kinematic Pseudophase & Energy Shaping Law:

Computes relative vertical and horizontal ankle-to-hip distances from thigh and shank IMU orientations at heel strike:

$z_{rel} = L_{thigh} \cdot \cos(\theta_{thigh}) + L_{shank} \cdot \cos(\theta_{shank})$

- Unified Virtual Spring & Damper Modulation: Acts as an upward assistive spring when step elevation difference indicates stair/ramp ascent, and modulates virtual spring-damper impedance via horizontal ankle distance during deep squat lifting and lowering.

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Participants: N = 10 healthy subjects performing realistic compound Logistics Lifting-Lowering-Carrying (LLC) circuits (squat lifting a weighted box, carrying across level ground, inclined ramps, and stairs, and lowering).
- Fatigue Protocol: Evaluated both pre-fatigue and immediately following an exhaustive quadriceps fatigue protocol, comparing Bare (No Exo) vs. Active Exoskeleton conditions.

7. Main Quantitative Results & Statistical Significance

- Quadriceps Effort Reduction: -14.5% holistic reduction across the complete LLC circuit ($p < 0.05$), reaching up to -28% during squat lifting and lowering.
- Post-Fatigue Performance Recovery: Reduced the fatigue-induced squat lifting completion-time penalty by 43%.
- Fatigue Mitigation & Stability: Attenuated median frequency (MDF) drop by 4 Hz and preserved trunk/knee postural stability.

8. Internal Study Limitations

1. Purely kinematic/FSR reactive controller: assistive gains remain fixed before and after muscular fatigue sets in.
2. Perceived exertion relied on subjective ratings; respiratory metabolic cost was not measured via indirect calorimetry.
3. Shoe insole FSRs cannot accurately isolate hand-held payload magnitude from dynamic body weight transfers.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

While Divekar et al. demonstrated that a QDD knee exoskeleton mitigates quadriceps fatigue, the controller itself is 'fatigue-blind'—sEMG is used only for offline evaluation. The proposed NMDT architecture incorporates real-time sEMG fatigue indices (MDF/RMS) directly into the closed-loop controller (Module C) to scale assistance as muscles tire.

10. Key Technical Discussions & Methodological Clarifications

Q1: Seamless Multi-Task Transition Without a Discrete Classifier:
Rather than relying on a finite-state machine or neural classifier prone to transition errors, the controller computes a continuous kinematic pseudophase variable from thigh and shank IMU orientations (relative ankle-to-hip spatial coordinates) gated by insole FSR contact forces, smoothly scaling virtual spring-damper torque across walking, stairs, and squatting.

Q2: Role of sEMG Signals During the Fatigue Protocol:
sEMG signals from the five thigh muscles (VMO, VL, RF, BF, ST) were recorded strictly for offline biomechanical evaluation to verify fatigue reduction; they were not fed into the real-time exoskeleton control loop.

Q3: Advantages of 9:1 Quasi-Direct Drive (T-Motor AK80-9) Actuation:
The low 9:1 planetary gear ratio yields minimal reflected rotor inertia and a backdrive torque below 2 Nm. This enables accurate open-loop current-based torque control without fragile series torque sensors and allows completely natural, uninhibited leg swing during carrying.

PAPER #3 (P3)

Q1 (SJR: 0.960) / Q2 (JCR IF: 3.4)

A Biomechanical Analysis of Weighted Lifting With an Active Knee Exoskeleton

Authors: Margaret A. Meagher, Grace R. Hunt, Brendon Ortolano, Rosemarie Murray, Lukas Gabert, Kenneth Bo Foreman, and Tommaso Lenzi (The University of Utah, USA) | **Year:** 2025 | **Journal:** IEEE Access, Vol. 13, pp. 135942–135953 | **DOI:** 10.1109/ACCESS.2025.3593446

SCImago (SJR): Q1 in Engineering (Misc.) & Computer Science (SJR = 0.960) | Clarivate Web of Science (JCR): Q2 (IF = 3.4)

1. Core Idea & Executive Overview

- Problem Statement: Repetitive weighted lifting causes rapid quadriceps fatigue, prompting workers to adopt poor lifting postures that increase knee and lumbar injury risk.
- Proposed Solution: The bilateral Utah Powered Knee Exoskeleton (55 Nm peak torque, 2.6 kg/leg) featuring a passive free-hinge ankle joint and a 3/4-length shoe footplate inserted into the footwear to transfer device weight and reaction forces directly to the ground without cuff migration. Controlled via a body-weight-normalized virtual torsional spring law, it achieves >53% reduction in biological knee extension moment during 9 kg box lifting.

2. Primary Research Question (RQ)

To what extent does an active knee exoskeleton governed by a linear virtual spring law reduce biological knee extension moments and quadriceps sEMG during 9 kg weighted squat lifting, and does it induce compensatory kinematic or kinetic penalties at the hip or ankle joints?

3. Hardware, Mechanical Design & Actuation Specifications

- Exoskeleton Platform: Bilateral Utah Powered Knee Exoskeleton — Mass: 2.6 kg per leg (including battery and interfaces).
- Actuation & Transmission: Maxon EC-22 120W BLDC motor coupled to a custom 3.8:1 gearbox architecture for high backdrivability.
- Torque Output: 55 Nm peak assistive torque (~21 Nm/kg torque density); Ingenia Everest Core 4-quadrant current driver; dual 32-bit PIC32 MCUs + Raspberry Pi 4 Compute Module.
- Grounding Interface: Passive free-hinge ankle joint with a 3/4-length shoe footplate that grounds vertical reaction forces.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs → Computed Outputs → System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: RLS RM08 magnetic commutation encoder + iC-Haus MU18S absolute joint output encoder (measures θ_knee).
- Laboratory Biomechanical Suite: 19-camera Vicon Vero 2.2 optical motion capture system (120 Hz), 2× Bertec force plates under the feet (1200 Hz), 1× AMTI OR6-7 force plate under the 9 kg box (1200 Hz), and Delsys Trigno Avanti wireless sEMG (1260 Hz) on bilateral Vastus Medialis (VM) and Vastus Lateralis (VL).

B) Computed / Predicted Outputs

- Active Controller Output: Target knee extension torque τ_{knee} computed from knee flexion angle via Eq. (1).
- Visual3D Biomechanical Outputs: Total inverse dynamics joint moments, net biological knee moment ($\tau_{bio} = \tau_{ID} - \tau_{exo}$), and peak/integrated VM and VL sEMG.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: Integrated absolute knee joint encoder (iC-Haus MU18S) ONLY.
- OFFLINE MONITORING & INVERSE DYNAMICS: 19 Vicon cameras, 3 force plates, and VM/VL sEMG are used strictly offline in Visual3D.

5. Control Architecture & Exact Mathematical Equations

- Eq. (1) — Virtual Torsional Spring Assistive Control Law (page 135944):

$$\tau_{knee} = \begin{cases} 0, & \text{if } \theta_{knee} < 15^\circ \text{ (deadzone for upright standing/walking)} \\ K \cdot (\theta_{knee} - \theta_{eq,knee}), & \text{if } \theta_{knee} > 15^\circ \text{ (active extension assistance)} \end{cases}$$

where $\theta_{eq,knee} = 15^\circ$ is the engagement threshold and K (Nm/deg) is a static stiffness normalized to subject body weight.

- Biological Knee Moment Computation in Visual3D:

$$\tau_{bio,knee} = \tau_{ID,total} - \tau_{knee,exo}$$

PAPER #3 (P3) — Margaret A. Meagher, Grace R. Hunt, Brendon Ortolano, Rosemarie Murray, Lukas Gabert, Kenneth Bo Foreman, and Tommaso Lenzi (2025) **Q1 (SJR: 0.960) / Q2 (JCR IF: 3.4)**

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Participants: N = 8 healthy adults performing symmetric squat lifting and lowering of a 9 kg box placed on the AMTI force plate.
- Conditions & Repetitions: 10–15 repetitions per condition across 3 randomized conditions: No Exoskeleton (NE), Transparent/No-Assist Mode (NA), and Active Exoskeleton Assist (EA). Knee flexion capped at ≤ 110°.

7. Main Quantitative Results & Statistical Significance

- Biological Knee Extension Moment Reduction: Peak biological knee moment decreased by 53.4% (left leg) and 53.58% (right leg) in Active Assist vs. No Exo ($p < 0.05$).
- Quadriceps sEMG Reduction: Peak sEMG of bilateral Vastus Medialis (VM) and Vastus Lateralis (VL) decreased by 29.2%, 31.2%, and 30.9% ($p < 0.05$).
- Adjacent Joint Preservation: No adverse compensatory changes in hip or ankle moments; footplate grounding maintained leg stability.

8. Internal Study Limitations

1. Purely position-dependent virtual spring $\tau = K(\theta - 15°)$: applies identical torque during eccentric lowering and concentric lifting regardless of movement velocity.
2. Stiffness K is normalized solely to subject body weight and does not adapt when external box weight varies (e.g., 0 kg vs. 9 kg vs. 25 kg).
3. Evaluated on N = 8 subjects in a constrained laboratory setting with a single 9 kg payload.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

A static spring law $K(\theta - 15°)$ cannot distinguish unweighted knee flexion from heavy lifting or differentiate descent from ascent. The NMDT architecture replaces this static law with a velocity- and load-aware Neural Torque Predictor (Module B) and sEMG Fatigue Modulator (Module C).

10. Key Technical Discussions & Methodological Clarifications

Q1: Extraction of Biological Knee Torque Under Exoskeleton Assistance:
Using 19 Vicon cameras and 2 Bertec force plates in Visual3D, bottom-up inverse dynamics computes the total net knee moment (biological + exoskeleton combined). Subtracting the measured exoskeleton actuator torque yields the true biological knee moment: $\tau_{bio} = \tau_{total_ID} - \tau_{exo}$, confirming a 53.5% biological reduction.

Q2: Purpose of the 15° Engagement Deadzone ($\theta_{eq,knee} = 15°$) in Equation (1):
Setting assistive torque to zero below 15° prevents the actuator from hyperextending or locking the knee joint during quiet standing and minor postural shifts.

Q3: Mechanical Role of the Free-Hinge Ankle and 3/4-Length Footplate:
Delivering 55 Nm at the knee generates substantial downward shear forces on thigh and shank straps. Routing a rigid frame to a passive hinged footplate inside the shoe transfers device weight and axial forces directly to the ground, preventing cuff migration.

PAPER #4 (P4)

Q2 (SJR: 0.730 / JCR Q2 IF: 3.0)

Controlling Negative and Positive Power for Efficiency Enhancement and Muscle Strain Mitigation During Squatting with a Portable Knee Exoskeleton

Authors: Shuangyue Yu, Lu Liu, Sainan Zhang, Antonio Di Lallo, Junxi Zhu, Qifei Wu, Guoyu Zuo, Xianlian Zhou, and Hao Su (North Carolina State University & Beijing University of Technology) | **Year:** 2025 | **Journal:** Annals of Biomedical Engineering, Vol. 53, pp. 1344–1358 | **DOI:** 10.1007/s10439-025-03696-0

SCImago (SJR): Q2 in Biomedical Engineering (SJR = 0.730) | Clarivate Web of Science (JCR): Q2 in Engineering, Biomedical (IF = 3.0)

1. Core Idea & Executive Overview

- Problem Statement: Squatting encompasses both eccentric descent (negative mechanical power) and concentric ascent (positive mechanical power). Conventional exoskeletons are either bulky/tethered or apply symmetric torque that over-assists during descent, inadvertently forcing hamstring antagonists to contract against the actuator.
- Proposed Solution: A lightweight 3.5 kg bilateral portable QDD knee exoskeleton governed by a 3-segment analytical biomechanical model. By systematically varying the eccentric-to-concentric assistance ratio $\beta \in \{0\%, 50\%, 100\%\}$, the study proves that moderate eccentric assistance ($\beta = 50\%$) is optimal, whereas full symmetric assistance ($\beta = 100\%$) triggers severe hamstring co-contraction.

2. Primary Research Question (RQ)

What is the optimal ratio of eccentric (negative power, squat-down) to concentric (positive power, stand-up) knee torque assistance in a portable QDD knee exoskeleton to minimize net metabolic energy expenditure, heart rate, and extensor/flexor muscle activation during repetitive squatting?

3. Hardware, Mechanical Design & Actuation Specifications

- Total System Mass: 3.5 kg bilateral portable knee exoskeleton (including waist control and battery pack).
- Actuator: Customized Myactuator RMD-X8 BLDC QDD motor (Mass: 630 g; Gear Ratio: 9:1 planetary; Nominal Torque: 18 Nm; Peak Torque: 54 Nm at 20.76 A; Backdrive Torque: 0.5 Nm; Reflected Inertia: 52.2 kg·cm^2; ROM: 0°–160°).
- Embedded Control & Power: Teensy 4.1 MCU running a 20 kHz current-loop PD controller (10.8 Hz bandwidth; sensorless current-based torque estimation); 22.2 V 2500 mAh LiPo battery (56 Wh, ~500 squats).

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs → Computed Outputs → System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: 4× wireless IMUs on bilateral thighs and shanks (measuring segment angles θ_{th}, θ_{sh} at 100 Hz) + internal magnetic motor encoders (measuring knee angle θ_K and velocity $\dot{\theta}_K$) + motor phase current.
- Physiological & Biomechanical Suite: Cosmed portable indirect calorimetry mask, Polar H10 heart-rate strap, 5-channel wireless sEMG (RF, VL, VM, BF, SEM), and Borg RPE exertion scale.

B) Computed / Predicted Outputs

- Active Controller Outputs: Real-time biological knee torque estimate τ_K from the 3-segment analytical model and phase-modulated assistive torque command τ_a.
- Evaluation Outputs: Net metabolic power (W/kg), Heart Rate (bpm), minute ventilation (L/min), Borg RPE, and extensor/flexor sEMG.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: 4× Wireless IMUs + Motor Magnetic Encoders + Motor Current (compute τ_K and phase transitions).
- OFFLINE MONITORING & EVALUATION: Cosmed calorimeter mask, Polar H10 HR monitor, 5-channel sEMG, and Borg RPE scale.

5. Control Architecture & Exact Mathematical Equations

- Eq. (2) & Eq. (3) (page 1349) — 3-Segment Analytical Knee Torque Model (Upper Body M_b, Thigh M_{th}, Shank M_{sh}):

Derived via Lagrangian mechanics using anthropometric mass/length fractions (Table 2) scaled by subject mass M_{sb} and height L_{sb}:

$\tau_K = f(\theta_K, \dot{\theta}_K, \ddot{\theta}_K, \theta_{th}, M_{sb}, L_{sb})$

[Methodological Note: In Figure 4's block diagram on page 1350, this block is labeled 'Torque Estimation (Eq. 9)' because intermediate Lagrangian derivations (Eqs. 2–8 in the draft) were moved to the Supplementary Material during publication.]

- Eq. (4) (page 1350) — Dual-Phase Eccentric/Concentric Assistive Control Law:

$$\tau_a = \begin{cases} \beta \cdot \alpha \cdot \tau_K, & \text{if } \theta_K < 95° \text{ AND } \dot{\theta}_K > 30°/s \text{ [Squat-Down / Negative Power Phase]} \\ \alpha \cdot \tau_K, & \text{otherwise [Stand-Up / Positive Power Phase]} \end{cases}$$

where $\alpha = 30\%$ is fixed concentric assistance and $\beta \in \{0\%, 50\%, 100\%\}$ scales eccentric assistance.

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Participants: N = 7 healthy adults performing repetitive paced bodyweight squats across 4 randomized conditions: (1) Baseline (No Exo), (2) β = 0% (Positive Power Only), (3) β = 50% (Moderate Negative + Positive Power), and (4) β = 100% (Full Symmetric Negative + Positive Power).

7. Main Quantitative Results & Statistical Significance

- Optimal Condition (β = 50% vs. Baseline — Table 3, page 1355):
  - Net Metabolic Cost reduced by -12.8% ($p < 0.05$) | Heart Rate reduced by -13.8% | Ventilation -8.9% | Borg RPE -1.8 points
  - Knee Extensor Mean sEMG reduced by: RF -43.2%, VL -42.1%, VM -39.4%
  - Knee Flexor Mean sEMG reduced by: BF -20.2%, SEM -18.9%
- Adverse Co-Contraction at β = 100%: Excessive upward torque during descent forced subjects to actively contract their hamstrings to flex the knee, increasing squat-down BF sEMG by +26.6% and SEM sEMG by +29.9%.

8. Internal Study Limitations

1. Evaluated strictly on unweighted bodyweight squats (0 kg external payload) under paced laboratory conditions.
2. Phase switching relies on fixed kinematic thresholds ($\theta_{Kset} = 95°$, $\dot{\theta}_{Kset} = 30°/s$).
3. Assistance gains (α = 30%, β = 50%) are static and do not adapt to progressive muscle fatigue.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Yu et al. established that eccentric descent assistance must be scaled to 50% of concentric assistance to prevent hamstring co-contraction, but their model assumes zero external payload. The NMDT architecture incorporates this 50% asymmetric eccentric scaling rule while dynamically estimating external box payloads and live sEMG muscle fatigue.

10. Key Technical Discussions & Methodological Clarifications

Q1: Clarification of 'Torque Estimation (Eq. 9)' in Figure 4 vs. Equations (2)–(3) in Text:

In the initial manuscript, the 3-segment Lagrangian derivation was expanded across 8 intermediate equations, making the final torque expression Eq. (9). During final typesetting, the intermediate steps were moved to the Supplementary Material and condensed into Eqs. (2)–(3) on page 1349, while the label inside Figure 4 remained '(Eq. 9)'.

Q2: Biomechanical Explanation for Why β = 50% Outperformed β = 100%:

During squat descent, gravity assists downward motion while quadriceps perform eccentric braking. At β = 50% (15% τ_K), the actuator shares the braking load comfortably. At β = 100% (30% τ_K), upward assistive torque opposes descent too strongly, forcing subjects to recruit their hamstring flexors (BF +26.6%, SEM +29.9%) to pull the body down against the motor.

Q3: Real-Time Phase Discrimination Between Squat-Down and Stand-Up (Eq. 4):

The controller monitors two encoder states simultaneously: knee flexion angle $\theta_K < 95°$ and flexion angular velocity $\dot{\theta}_K > 30°/s$. When both conditions hold, the user is actively descending (applying $\beta \cdot \alpha \cdot \tau_K$); once $\theta_K \geq 95°$ at the bottom turnaround or velocity reverses, full concentric assistance ($\alpha \cdot \tau_K$) is engaged.

PAPER #5 (P5)

Q1 (SJR: 1.150 / JCR Q1 IF: 4.3)

Development of a Training-Oriented Wearable Knee Joint Exoskeleton for Forming a Scientific Force Application Pattern in Squat Tasks

Authors: Shuai Chang, Feng Feng, Zihao Li, and Yang Yu (Capital University of Physical Education and Sports & Capital Normal University, Beijing, China) | **Year:** 2026 | **Journal:** Frontiers in Bioengineering and Biotechnology, Vol. 14, Art. 1719023 | **DOI:** 10.3389/fbioe.2026.1719023

SCImago (SJR): Q1 in Biotechnology & Bioengineering (SJR = 1.150) | Clarivate Web of Science (JCR): Q1 (IF = 4.3)

1. Core Idea & Executive Overview

- Problem Statement: Prolonged reliance on assistive exoskeletons can diminish intrinsic muscular activation in athletes and trainees, while conventional damping-shaft resistive braces suffer from non-linear torque adjustment and cuff slippage.
- Proposed Solution: A training-oriented clutch-disc wearable knee exoskeleton that applies stepless rotational resistance (14, 28, and 42 Nm) to inhibit excessive knee loading and guide the wearer toward a safer 'hip-dominant' squat strategy.

2. Primary Research Question (RQ)

Can a clutch-disc-based wearable resistive knee exoskeleton redistribute lower-limb joint power from a knee-dominant pattern to a hip-dominant pattern compared to mechanically work-matched barbell squats, and how does joint resistance affect lower-limb kinematics and muscle activation?

3. Hardware, Mechanical Design & Actuation Specifications

- Resistance Mechanism: Custom clutch-disc module with alternating brass and steel friction discs compressed between internal and external pressure discs by four helical springs.
- Two Structural Variants (Figs. 3 & 4): (1) Manual lead-screw adjustment prototype (3D-printed resin links + CNC-machined metal clutch, deployed in human trials), and (2) Motor-driven rack-and-pinion stepless adjustment prototype.
- Calibrated Torque Levels: Unilateral lever arm r = 0.20 m with normal forces F ∈ {70, 140, 210} N yielding exact resistive torques of 14 Nm, 28 Nm, and 42 Nm.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: None during human trials (open-loop pre-calibrated mechanical clutch resistance settings).
- Laboratory Biomechanical Suite: 3D infrared optical motion capture system (plug-in-gait marker set including C7), dual 3D ground reaction force plates, and 5-channel wireless sEMG (VM, RF, TA, GM, BF).

B) Computed / Predicted Outputs

- OpenSim & MATLAB Biomechanical Outputs: 3D hip, knee, and ankle joint angles/ROM, net joint moments, joint mechanical power contributions (%), and normalized %MVC sEMG activation.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: None (pre-set mechanical clutch friction).
- OFFLINE MONITORING & BIOMECHANICAL MODELING: 100% of the sensor suite (MoCap, dual force plates, 5-ch sEMG, OpenSim) was utilized offline for biomechanical evaluation.

5. Control Architecture & Exact Mathematical Equations

- Equivalent Gravitational Load & Unilateral Clutch Torque (page 5):

$G = m \cdot g$ (70 N ≈ 7.14 kg, 140 N ≈ 14.29 kg, 210 N ≈ 21.43 kg) | $\tau = r \times F$ ($r = 0.20$ m ⇒ $\tau \in \{14, 28, 42\}$ Nm)

- Biomechanical Equivalence (Work-Matching) Protocol (pages 5–6):

$W = F \times s \times \cos(\theta)$

  - Barbell Control Group: $F_{barbell} = m_{barbell} \cdot g$, vertical C7 displacement $s_{barbell} = 0.42 \pm 0.03$ m.
  - Exoskeleton Group: Equivalent force $F_{exo} = \tau / r_{knee}$ ($r_{knee} = 0.18 \pm 0.02$ m), arc displacement $s_{exo} = \theta_{knee} \times r_{knee}$ ($\theta_{knee} = 1.8$ rad).
  - Pilot Calibration (N = 3 at 28 Nm vs. 280 N): Confirmed no difference in peak GRF (1256 ± 89 N vs. 1238 ± 94 N, $p > 0.05$) or total mechanical work (387 ± 42 J vs. 392 ± 38 J, $p > 0.05$).

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Participants: N = 15 healthy male athletes from Capital University of Physical Education and Sports (Age: 20.4 ± 1.0 yrs, Mass: 80.1 ± 4.7 kg, Height: 179.0 ± 4.5 cm; sample size determined via G*Power 3.1.9.7 repeated-measures ANOVA, f = 0.8).
- Conditions: Exoskeleton resistance squats (14, 28, 42 Nm) vs. work-matched free-weight barbell squats.

7. Main Quantitative Results & Statistical Significance

- Joint Power Redistribution: Hip power contribution increased by +20% to +40% ($p < 0.05$), while knee power contribution decreased by -20% to -30% ($p < 0.05$), successfully inducing a hip-dominant movement strategy.
- Muscle Activation (sEMG): Vastus Medialis (VM) activation decreased by ~50% ($p < 0.05$) compared to barbell squats.
- Kinematic Trade-Off: Uncompensated joint resistance and cuff constraints reduced peak knee and ankle flexion ROM by 7° to 9° ($p < 0.05$).

8. Internal Study Limitations

1. Designed as a resistive sports-training brace rather than an active occupational lifting-assistance exoskeleton.
2. Resistive friction torque remains constant across the squat cycle rather than adapting to joint angle or velocity.
3. Imposes a 7°–9° loss in knee and ankle flexion ROM, which restricts deep squat postures.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Chang et al. quantified that uncompensated mechanical impedance at the knee restricts knee and ankle ROM by 7°–9° and alters natural inter-joint coordination. The NMDT architecture employs low-impedance QDD actuation with active friction compensation to preserve full natural deep-squat ROM.

10. Key Technical Discussions & Methodological Clarifications

Q1: Relevance of a Resistive Training Exoskeleton to Our NMDT Design:

First, its OpenSim + MoCap + Force Plate methodology demonstrates how external knee torque redistributes hip-versus-knee mechanical power (+20–40% hip shift). Second, it quantifies the exact kinematic penalty (7°–9° ROM loss) caused by uncompensated knee impedance, justifying active transparency control in our design.

Q2: Mechanical Work-Matching Protocol Between Knee Torque (Nm) and Shoulder Barbell Load (kg):

Using $W = F \cdot s \cdot \cos\theta$, the rotational work at the knee joint (effective moment arm $r_{knee} = 0.18$ m, angular excursion $\theta = 1.8$ rad) was equated to the vertical gravitational work of the barbell measured at the C7 vertebra ($s = 0.42$ m), and verified on 3 pilot subjects (387 J vs. 392 J, $p > 0.05$).

Q3: Advantages of the Alternating Brass-and-Steel Clutch Disc Mechanism:

Conventional damping shafts exhibit non-linear nut-tension-to-torque curves and limited adjustment range. Compressing alternating brass and steel friction plates via four springs yields linear, stepless friction torque regulation across 14, 28, and 42 Nm.

PAPER #6 (P6) **Q1 (SJR: 0.850) / Q2 (JCR IF: 3.7)**

Squat Training Assisted by a Rigid-Soft Hybrid Knee Exoskeleton: A Preliminary Study

Authors: Xiaolin Dai, Ming Xu, Heran Zhong, Zhihao Zhou, Rongli Wang, Xuewen Rong, Yibin Li, and Qining Wang (Peking University & Shandong University, China) | **Year:** 2025 | **Journal:** IEEE Transactions on Medical Robotics and Bionics (TMRB), Vol. 7, No. 4, pp. 1565–1575 | **DOI:** 10.1109/TMRB.2025.3604140

SCImago (SJR): Q1 in Biomedical Engineering & Rehabilitation (SJR = 0.850) | Clarivate Web of Science (JCR): Q2 (IF = 3.7)

1. Core Idea & Executive Overview

- Problem Statement: High tibiofemoral joint contact forces during squatting exacerbate knee cartilage wear and pain, while mounting heavy motors directly on the knee increases distal limb inertia.
- Proposed Solution: A lightweight (1.2 kg on-body) unilateral rigid-soft hybrid knee exoskeleton actuated via bidirectional Bowden cables from an off-board motor unit and governed by a 3-level Assist-As-Needed (AAN) admittance controller with a 'Virtual Tunnel'. Evaluated during a 30-second isometric 60° squat hold using OpenSim and a Calibrated EMG-Informed Neuromusculoskeletal (CEINMS) model, achieving a 29.0% reduction in internal knee joint contact force.

2. Primary Research Question (RQ)

How does a rigid-soft hybrid knee exoskeleton governed by a virtual-tunnel Assist-As-Needed (AAN) admittance controller affect internal tibiofemoral joint contact forces, individual muscle forces (via CEINMS), and multi-muscle synergies during sustained isometric squatting?

3. Hardware, Mechanical Design & Actuation Specifications

- Wearable On-Body Unit: Unilateral rigid-soft hybrid knee brace with a cable-driven linear actuator — Mass: 1.2 kg on-body; Peak Linear Force: 260 N; Max Linear Velocity: 1.2 m/s; Stroke: 110–170 mm.
- Off-Board Actuation Bench (via Bowden cables): Brushless DC motor + planetary reducer (10 Nm rated, 20 Nm peak, 300 rpm), 18-bit Heidenhain optical encoder, Elmo Twitter servo driver (25A/100V), STM32H743 MCU (100 Hz), and 48 V 3600 mAh LiPo.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: 2× IMUs on thigh and shank (relative orientation yields knee flexion angle θ_k), 18-bit Heidenhain motor optical encoder (measures linear cable displacement X), and a Sunrise Instruments 6-axis force sensor mounted between the linear actuator and shank cuff (measures distal assistive force F_a directly, eliminating upstream Bowden-cable friction error).
- Laboratory Biomechanical Suite: 16-marker Vicon optical motion capture (100 Hz), 2× Bertec force plates (1000 Hz), and 16-channel Delsys Trigno wireless sEMG (2000 Hz) on 8 right-leg muscles (RF, VL, VM, BF, ST, TA, MG, LG).

B) Computed / Predicted Outputs

- Active Controller Outputs (3 Hierarchical Levels): High-level FSM state (Stand vs. Squat at 5° threshold), mid-level AAN admittance desired displacement X_r, and low-level motor position tracking.
- OpenSim + CEINMS Outputs: Net knee moment, internal tibiofemoral joint contact force (JCF), muscle-tendon forces, and NNMF muscle synergies.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: 2× Thigh/Shank IMUs + 18-bit Heidenhain Encoder + Sunrise 6-Axis Force Sensor.
- OFFLINE MONITORING & NEUROMUSCULOSKELETAL MODELING: Vicon MoCap, Bertec force plates, and 8-muscle sEMG are used strictly offline to drive OpenSim Gait2392, CEINMS, and NNMF muscle synergy analysis.

5. Control Architecture & Exact Mathematical Equations

- Eq. (1) & Eq. (2) (page 1567): Actuator length $L = a\cdot\theta_k^2 + b\cdot\theta_k + c$ | Assistive torque $\tau_a = F_a \cdot H(\theta_k)$
- High-Level FSM: Transitions from Stand State to Squat State when $\theta_k > 5°$; returns to Stand when $\theta_k < 5°$ (zero force at initial position X_{ri}).
- Eq. (3)–(8) — Mid-Level AAN Admittance Controller with Virtual Tunnel:
  - Virtual Tunnel Error (Eq. 3): $\theta = \{ 0, \text{if } |\theta_e| \le \theta_{bound} ; \theta_e \pm \theta_{bound}, \text{if } |\theta_e| > \theta_{bound} \}$
  - Admittance Law (Eq. 4–8): $M_d\ddot{X} + D_d\dot{X} + K_dX = F_e$ with $M_d = 0$, $K_d = \min(K, K_{max})$ ($K = K_b$ inside tunnel, $K = c_k|\theta|$ outside tunnel), and critical damping $D_d = 0.5\sqrt{K_d}$. (Reduced unnecessary actuator adjustments by 80%; mean force = 78.71 ± 1.80 N).

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Participants: N = 7 healthy young males (Age: 24.9 ± 2.5 yrs, Height: 175.4 ± 4.1 cm, Mass: 69.1 ± 5.8 kg).
- Task & Timing (0 kg External Payload): 32-second squat cycle across 3 phases: (1) 1 s descent to $\theta_k = 60°$, (2) 30 s isometric hold at 60° (with visual screen feedback), and (3) 1 s ascent to standing, followed by 30 s rest (6 sets: Exo-On vs. Exo-Off).

7. Main Quantitative Results & Statistical Significance

- Joint & Muscle Force Reductions (Exo-On vs. Exo-Off):
  - Internal Tibiofemoral Knee Joint Contact Force (JCF): -29.00% ($p < 0.05$)
  - Net Knee Joint Extension Moment: -14.53% ($p < 0.05$) | Net Quadriceps Muscle Force (CEINMS): -24.40% ($p < 0.01$)
  - Net 8-Muscle sEMG Activation: -23.48% ($p < 0.01$) (Individual: RF -31.23%, ST -21.84%, VM -19.06%, VL -17.18%).

8. Internal Study Limitations

1. Unilateral prototype tethered via Bowden cables to an off-board actuator bench; evaluated on N = 7 healthy young males.
2. Tested only during static/isometric 60° squat holding (30 s) with zero external payload (0 kg).
3. CEINMS neuromusculoskeletal modeling is executed offline rather than inside the real-time control loop.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Dai et al. demonstrated the value of EMG-informed neuromusculoskeletal modeling (CEINMS) for quantifying internal knee joint contact force reductions (-29%), but ran it offline. The proposed NMDT architecture embeds a real-time Musculoskeletal Digital Twin (Module A + Module C) inside the active control loop during dynamic weighted lifting.

10. Key Technical Discussions & Methodological Clarifications

Q1: Squat Phase Breakdown and Role of the 5° FSM Threshold:

Each squat trial comprises three phases: a 1 s descent to 60° knee flexion, a 30 s isometric hold at 60°, and a 1 s ascent. The high-level FSM monitors the thigh/shank IMUs: when $\theta_k < 5°$, the controller remains in the Stand State (locking cable position at X_{ri} with zero assistive force); once $\theta_k > 5°$, it engages the Squat State and activates the mid-level AAN admittance controller.

Q2: Mitigation of Non-Linear Bowden-Cable Friction Error:

Rather than inferring assistive force from upstream motor current on the bench, the authors mounted a Sunrise Instruments 6-axis force sensor directly at the distal attachment between the linear actuator and the shank cuff, measuring true human–exoskeleton interaction force F_a.

Q3: Function of the Virtual Tunnel and Zero Virtual Mass ($M_d = 0$) in Equations (3)–(8):

Within the virtual tunnel ($|\theta_e| \leq \theta_{bound}$), the controller maintains a low baseline stiffness K_b so natural postural micro-sways are uninhibited, reducing unnecessary motor adjustments by 80%. When fatigue causes deviation outside the tunnel, stiffness increases linearly ($K = c_k|\theta|$). Setting virtual mass $M_d = 0$ reduces the admittance filter to a stable first-order spring-damper law, preventing high-frequency oscillations.

PAPER #7 (P7)

Q1 (SJR: 1.560 / JCR Q1) — Both Journals Verified Q1

The Only Way Is Up: Active Knee Exoskeleton Reduces Muscular Effort in Quadriceps During Weighted Stair Ascent (Boon et al., 2025) | Active Knee Joint Exoskeleton for Stair Ascent Augmentation (Zhang et al., 2021)

Authors: Vincent S. Boon, Brendon Ortolano, Andrew J. Gunnell, Margaret Meagher, Rosemarie C. Murray, Lukas Gabert, and Tommaso Lenzi (2025, Univ. of Utah) | Zongwei Zhang, Jizhuang Fan, Hongzhe Jin, Tianjiao Zheng, Sikai Zhao, Shun Ma, Jie Zhao, and Yanhe Zhu (2021, Harbin Inst. of Tech.) | **Year:** 2025 (RA-L) / 2021 (SCIS) | **Journal:** IEEE Robotics and Automation Letters (RA-L), Vol. 10, Issue 5, pp. 4484–4491 | Science China Information Sciences, Vol. 64, No. 3, Art. 139204 | **DOI:** 10.1109/LRA.2025.3551543 | 10.1007/s11432-018-9767-6

IEEE RA-L: Q1 in Robotics (SJR = 1.560) | Science China Information Sciences: Q1 in Computer Science & Information Systems (IF = 7.6)

1. Core Idea & Executive Overview

- Problem Statement: Loaded stair ascent demands substantial concentric knee extension power against gravity, causing rapid quadriceps fatigue and high patellofemoral loading. Both benchmark stair-ascent studies are documented below:
  - [A] Boon et al. (2025, IEEE RA-L): Evaluates the bilateral Utah Powered Knee Exoskeleton (2.6 kg/leg, 55 Nm peak) on N = 8 healthy adults ascending stairs with a 9.1 kg (20 lb) weighted vest using a Finite-State Machine (FSM) controller.
  - [B] Zhang et al. (2021, Sci. China Inf. Sci.): Develops a ball-screw + 3-hinge crank-slider knee exoskeleton (2.9 kg leg weight, 180 Nm peak torque at 95° flexion) with a 5-link gravity-compensation stance controller and Lyapunov-stable switching law, tested during stair ascent with a 30 kg backpack load.

2. Primary Research Question (RQ)

- [Boon et al., 2025]: How effectively does an active knee exoskeleton governed by a stance/swing FSM reduce quadriceps sEMG during loaded stair ascent (9.1 kg vest) while preserving natural stair kinematics?
- [Zhang et al., 2021]: How can a ball-screw crank-slider knee exoskeleton combine 5-link stance gravity compensation with swing zero-force impedance via a Lyapunov switching controller to compensate 80% of knee torque under a 30 kg load?

3. Hardware, Mechanical Design & Actuation Specifications

- [A] Boon et al. (2025): Bilateral Utah Powered Knee Exoskeleton — 2.6 kg/leg, Maxon EC-22 120W BLDC motor, 3.8:1 gearbox, 55 Nm peak torque, passive free-hinge ankle with 3/4-length shoe footplate.
- [B] Zhang et al. (2021): Linear ball-screw actuator driving a 3-hinge crank-slider linkage (mimicking biological patella moment-arm variation to deliver 180 Nm peak torque at 95° flexion); 2.9 kg leg unit (13 kg total with battery backpack); ROM: -3° to 125°.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs → Computed Outputs → System Utilization

A) Sensory Inputs (Onboard & Lab)

- [A] Boon et al. (2025): Onboard absolute knee encoder + shank/thigh IMU for FSM stance/swing transitions (Active Control); 8-channel wireless sEMG on quadriceps (VM, VL, RF) and hamstrings (BF) (Offline Evaluation).
- [B] Zhang et al. (2021): Absolute knee encoder at hinge O1 + 2× multidimensional interaction force sensors on thigh and shank + plantar pressure sensor insoles under both feet (100% Active Real-Time Control).

B) Computed / Predicted Outputs

- [A] Boon et al. (2025): Stance-phase knee extension torque profile; peak and integrated quadriceps sEMG reductions.
- [B] Zhang et al. (2021): Stance gravity-compensation torque $\hat{T}_{K1} = 0.8 \cdot T_{K1}$ (80% offloading), swing zero-force impedance ($|F_{int}| \leq 15$ N), and Lyapunov switching control output u_s eliminating transition torque spikes within < 0.2 s.

C) Output Utilization (Active Control vs. Monitoring)

- [A] Boon et al. (2025): Joint Encoder + IMU = ACTIVE CONTROL | 8-ch sEMG = OFFLINE MONITORING.
- [B] Zhang et al. (2021): All onboard sensors (Knee Encoder + 2× Multi-Axis Force Sensors + Pressure Insoles) = 100% ACTIVE CONTROL.

5. Control Architecture & Exact Mathematical Equations

- [A] Boon et al. (2025): FSM Stance-Extension / Swing-Transparency torque law gated by heel-strike and knee extension velocity.
- [B] Zhang et al. (2021) Complete Equations (Pages 1–2, Eqs. 1–18):
  - Eq. (1): Non-linear crank-slider transmission mapping ball-screw thrust to knee torque (180 Nm peak at 95°).
  - Eq. (2)–(4): 5-link sagittal inverted-pendulum gravity compensation in stance: $\hat{T}_{K1} = \eta \cdot T_{K1}$ ($\eta = 80\%$).
  - Eq. (5)–(6): Swing-phase zero-force impedance controller: $M_d\ddot{\theta} + B_d\dot{\theta} + K_d(\theta - \theta_d) = F_{int}$ (maintaining $|F_{int}| \leq 15$ N).
  - Eq. (7)–(18): State-space Lyapunov switching law ($F^T P + P F = -2Q$, $V(x) = x^T P x$) ensuring smooth stance-swing transition in < 0.2 s.

PAPER #7 (P7) — Vincent S. Boon, Brendon Ortolano, Andrew J. Gunnell, Margaret Meagher, Rosemarie C. Murray, Lukas Gabert, and Tommaso Lenzi (2025 (RA-L) / 2021 (SCIS)) **Q1 (SJR: 1.560 / JCR Q1) — Both Journals Verified Q1**

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- [A] Boon et al. (2025): N = 8 healthy adults ascending an instrumented staircase wearing a 9.1 kg (20 lb) weighted vest across No Exoskeleton, Transparent Mode, and Active Assist conditions.
- [B] Zhang et al. (2021): 3 stair-climber experiments including 10-minute stair ascent at 10 steps/min carrying a 30 kg backpack load (3 repetitions with vs. without exoskeleton; 2 hr rest intervals), recording heart rate and pulmonary ventilation.

7. Main Quantitative Results & Statistical Significance

- [A] Boon et al. (2025): Reduced peak quadriceps sEMG by -32% and integrated (mean) quadriceps sEMG by -29% ($p < 0.05$) during 9.1 kg weighted stair ascent without altering natural cadence.
- [B] Zhang et al. (2021): Compensated 80% of required biological knee torque during 30 kg stair ascent (comfortable flexion up to 85°); Lyapunov switching eliminated torque impulses in < 0.2 s and significantly lowered heart rate and ventilation.

8. Internal Study Limitations

1. Boon et al. (2025) uses fixed FSM parameters for a constant 9.1 kg vest without adapting to progressive muscle fatigue.
2. Zhang et al. (2021) has a high total system mass (13 kg with backpack), is specialized for stair ascent only, and assumes known payload mass.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Both stair-ascent controllers assume a known static payload (9.1 kg or 30 kg) and do not adapt to progressive muscle fatigue. The NMDT architecture combines smooth state transitions with online payload/torque prediction and live sEMG fatigue scaling.

10. Key Technical Discussions & Methodological Clarifications

Q1: Hybrid Stance/Swing Control and Lyapunov Switching in Zhang et al. (2021):

During stance (detected by plantar pressure insoles), the controller executes a 5-link gravity-compensation law (Eqs. 2–4) providing 80% of required knee extension torque. During swing, it transitions via a Lyapunov-stable switching law (Eqs. 7–18) to a zero-force impedance controller (Eqs. 5–6) using dual thigh/shank interaction force sensors to keep resistive force within ±15 N.

Q2: Biomimetic Advantage of the 3-Hinge Crank-Slider Linkage in Zhang et al. (2021):

Loaded stair ascent requires peak knee extension torque at deep flexion (~85°–95°). The 3-hinge crank-slider linkage emulates the human patella by increasing the effective moment arm with flexion angle, achieving 180 Nm peak torque precisely at 95°.

Q3: Complementary Findings of Boon et al. (2025, IEEE RA-L):

While Zhang et al. (2021) demonstrated high-load (30 kg) mechanical compensation and cardiorespiratory savings, Boon et al. (2025) conducted a multi-subject (N = 8) electromyography investigation using the lighter Utah Knee Exoskeleton (2.6 kg/leg, 9.1 kg vest), confirming a 32% peak and 29% mean reduction in quadriceps sEMG.

PAPER #8 (P8)

Q1 (SJR: 0.810) / Q2 (JCR IF: 3.4)

Optimization of Torque-Control Model for Quasi-Direct-Drive Knee Exoskeleton Robots Based on Regression Forecasting

Authors: Yuxuan Xia, Wei Wei, Xichuan Lin, and Jiaqian Li (Soochow University & Suzhou Institute of Biomedical Engineering and Technology, Chinese Academy of Sciences, China) | **Year:** 2024 | **Journal:** Sensors (MDPI), Vol. 24, Issue 5, Art. 1505 | **DOI:** 10.3390/s24051505

SCImago (SJR): Q1 in Instrumentation & Electrical Engineering (SJR = 0.810) | Clarivate Web of Science (JCR): Q2 (IF = 3.4)

1. Core Idea & Executive Overview

- Problem Statement: Despite the high responsiveness of Quasi-Direct-Drive (QDD) actuators, the cumulative latency of IMU sampling, neural inference, CAN-bus communication, and PID current settling introduces an inherent ~50 ms phase lag—causing assistive torque to lag behind peak biological muscle demand.
- Proposed Solution: The bilateral carbon-fiber MebotX knee exoskeleton (1.8 kg/leg, 24.8 Nm peak) driven by a hybrid LeNet CNN + LSTM regression forecasting model receiving a 48×48 spatiotemporal feature matrix from 6 wearable IMUs (200 Hz). By introducing a 50 ms forward time-shifted reference (t_s = 50 ms), actuator phase delay is compensated, achieving a tracking RMSE of 4.64 × 10^{-3} Nm/kg and a 19.37% mean quadriceps sEMG reduction during stair ascent.

2. Primary Research Question (RQ)

Can a hybrid CNN-LSTM regression forecasting network driven by 6 wearable IMUs, combined with a 50 ms forward time-shifted torque reference (t_s = 50 ms), eliminate electromechanical control lag in a QDD knee exoskeleton and reduce quadriceps muscular activation during stair climbing?

3. Hardware, Mechanical Design & Actuation Specifications

- Exoskeleton Platform: MebotX bilateral carbon-fiber knee exoskeleton — Mass: 1.8 kg per leg; ROM: -5° to 160°.
- Custom QDD Actuator: Mass: 521 g; Dimensions: Φ89 × 50.25 mm; 10:1 planetary reducer; Peak Torque: 24.8 Nm (2.48 Nm BLDC motor); Reflected Inertia: 32.2 kg·cm^2; 24 V power supply.
- Embedded Electronics: 14-bit AS5048 magnetic encoder + STM32F405RGT6 CAN-bus motor driver.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: 6× wearable IMUs (YIS106-V, 200 Hz sampling rate, mounted bilaterally on thighs, calves, and feet; 6 IMUs × 8 channels over a 48-sample sliding window = 48×48 spatiotemporal input matrix X) + 2× onboard joint torque sensors + 14-bit AS5048 magnetic encoders.
- Offline Ground-Truth Training & Evaluation Suite: 8-camera Vicon MX optical motion capture + Kistler 3D force plates integrated into a 20 cm staircase (used offline to generate inverse dynamics ground-truth knee torque labels), plus Ambu Ag/AgCl surface sEMG on Rectus Femoris (RF) and Vastus Medialis (VM).

B) Computed / Predicted Outputs

- Active Controller Outputs: Time-shifted predicted biological knee torque $\tau_ref(t + 50\ ms)$ tracked via closed-loop PID.
- Evaluation Outputs: Torque tracking RMSE (Nm/kg) and maximum RMS sEMG of RF and VM across $t_s \in \{0, 20, 50\}$ ms.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: 6× Wearable IMUs (200 Hz) + 2× Actuator Torque Sensors + AS5048 Magnetic Encoders.
- OFFLINE GROUND-TRUTH & EVALUATION: 8-camera Vicon MX + Kistler force plates (training label collection only) and Ambu sEMG on RF & VM (experimental evaluation only).

5. Control Architecture & Exact Mathematical Equations

- Spatiotemporal Input Matrix Construction: 6 IMUs × 8 channels = 48 spatial channels × 48 time samples ⇒ $X \in R^{48\times48}$.
- Hybrid LeNet CNN + LSTM Architecture (Eqs. 1–6, pages 6–8):
  - 3 Convolutional layers + 2 Max-Pooling layers + 2 Fully Connected layers extract a 1×3000 spatial feature vector.
  - LSTM gates (Forget f_t, Input i_t, Cell state C_t, Output o_t) model temporal gait dynamics.
- Time-Shifted Reference Lead Compensation: $\tau_cmd(t) = \alpha \cdot \hat{\tau}_knee(t + t_s)$ with optimal lead shift t_s = 50 ms.

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Participants: N = 5 healthy adults (4 males, 1 female) ascending a 12-step staircase (step height: 20 cm).
- Experimental Conditions: NO-EXO, EXO-OFF (unpowered exoskeleton), and Active Assist across 3 lead-time shifts: t_s = 0 ms, t_s = 20 ms, and t_s = 50 ms.

7. Main Quantitative Results & Statistical Significance

- Torque Tracking Accuracy: At the optimal time shift t_s = 50 ms, torque tracking RMSE reached 4.64 × 10^{-3} Nm/kg.
- Quadriceps sEMG Reductions (t_s = 50 ms vs. NO-EXO — Table 2, page 14 & Section 4, page 15):
  - Rectus Femoris (RF) max RMS sEMG decreased by -20.87%.
  - Vastus Medialis (VM) max RMS sEMG decreased by -17.45% (Combined average quadriceps reduction: -19.37%).

[Methodological Note on Abstract Discrepancy: The Abstract contains a typographical error stating 'VM increased by 17.45%'; Table 2 (page 14), Figure 11, and Section 4 (page 15) verify that VM DECREASED by 17.45% at t_s = 50 ms, whereas wearing the unpowered exoskeleton (EXO-OFF) increased VM by +8.57%.]

8. Internal Study Limitations

1. Requires strapping 6 separate IMU modules across both thighs, shanks, and feet.
2. Trained on unloaded stair ascent only; purely kinematic IMU inputs cannot detect external hand-held payload mass.
3. The forward time shift t_s = 50 ms is fixed rather than adapting to cadence variations.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Xia et al. demonstrated that CNN-LSTM regression with a 50 ms time-shifted reference eliminates QDD actuator lag, but a purely IMU-driven network is blind to external payloads and muscular fatigue. Our NMDT architecture upgrades Module B with kinetic load features and couples it with active sEMG fatigue modulation (Module C).

10. Key Technical Discussions & Methodological Clarifications

Q1: Verification of Vastus Medialis (VM) sEMG Reduction in Table 2 vs. Abstract Typo:

While the Abstract contains a wording error ('VM increased by 17.45%'), Section 3.3 (Table 2, page 14), Figure 11, and Section 4 (page 15) explicitly show that unpowered EXO-OFF increased VM activation by +8.57%, whereas active assistance progressively reduced VM activation by -14.0% (t_s = 0 ms), -16.5% (t_s = 20 ms), and -17.45% (t_s = 50 ms), yielding an overall 19.37% quadriceps reduction.

Q2: Formulation of the 48×48 Spatiotemporal Input Matrix:

Each of the 6 IMUs provides 8 kinematic signals (triaxial acceleration, triaxial angular velocity, roll, and pitch), forming 6 × 8 = 48 spatial channels. Stacking 48 consecutive time samples (at 200 Hz = 240 ms sliding window) produces a 48×48 matrix suitable for 2D LeNet spatial feature extraction prior to LSTM temporal sequence modeling.

Q3: Necessity of the 50 ms Forward Time Shift in a QDD System:

Although QDD motors exhibit low mechanical inertia, the end-to-end control pipeline (IMU acquisition, CNN-LSTM inference, CAN-bus packet transmission, and PID torque settling) introduces ~50 ms of phase lag. Advancing the reference trajectory by t_s = 50 ms synchronizes physical motor peak torque with human muscle demand.

PAPER #9 (P9)

Q1 (SJR: 2.540 / JCR Q1 IF: 6.1)

Robust Adaptive Neural Force Control of a SEA-Based Knee Exoskeleton With Low Impedance

Authors: Yanwei Zhao, Songhao Li, Jianda Han, and Weiguang Huo (College of Artificial Intelligence, Nankai University, Tianjin, China) | **Year:** 2025 | **Journal:** IEEE/ASME Transactions on Mechatronics (TMECH) | **DOI:** 10.1109/TMECH.2025.3569854

SCImago (SJR): Q1 in Mechanical & Control Engineering (SJR = 2.540) | Clarivate Web of Science (JCR): Q1 (IF = 6.1)

1. Core Idea & Executive Overview

- Problem Statement: Non-linear linkage kinematics, actuator friction, and unmodeled human–exoskeleton interaction disturbances degrade closed-loop force tracking and increase parasitic impedance in Series Elastic Actuators (SEAs).
- Proposed Solution: A Crank-Slider Series Elastic Actuator (CS-SEA) knee exoskeleton paired with a Robust Adaptive Neural Network Force Controller (RANNFC) that approximates unmodeled dynamics and friction online, achieving high-precision torque tracking (0.31 Nm RMSE) and low transparent-mode resistive torque (1.03 Nm).

2. Primary Research Question (RQ)

How can a Crank-Slider Series Elastic Actuator (CS-SEA) governed by a Robust Adaptive Neural Network Force Controller (RANNFC) achieve sub-Newton-meter closed-loop torque tracking accuracy and low mechanical impedance across non-linear knee trajectories?

3. Hardware, Mechanical Design & Actuation Specifications

- Actuator Topology: Crank-Slider Series Elastic Actuator (CS-SEA) integrating a physical elastic spring element in series with a slider-crank transmission to match the variable moment-arm profile of the human knee joint.
- Mechanical Characteristics: Compliant series spring absorbs impact shocks and converts force control into spring-deflection regulation; slider-crank geometry provides non-linear kinematic gearing.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control Loop: Dual high-resolution optical encoders (measuring motor crank input angle and knee joint output angle) + Series Elastic spring deflection sensor measuring real-time interaction torque via Hooke's law ($\tau = K_s \cdot \Delta\theta$).

B) Computed / Predicted Outputs

- Active Controller Outputs: Online adaptive neural network friction/disturbance compensation term and closed-loop motor drive command regulating spring deflection to track desired assistive or zero-impedance torque.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: 100% of onboard sensors (Dual Optical Encoders + Spring Deflection Sensor) operate inside the real-time RANNFC closed-loop force controller.
- BIOLOGICAL MONITORING: None (electromechanical trajectory tracking and transparent-mode impedance validation).

5. Control Architecture & Exact Mathematical Equations

- Series Elastic Torque Law: $\tau_{actual} = K_{spring} \cdot (\theta_{crank}(x_{slider}) - \theta_{knee})$
- RANNFC Control Law: Radial Basis Function (RBF) adaptive neural weights $\hat{W}$ approximate non-linear crank-slider kinematics, backlash, and Coulomb/viscous friction online, augmented by a robust sliding-mode term guaranteeing Lyapunov stability and bounded tracking error.

PAPER #9 (P9) — Yanwei Zhao, Songhao Li, Jianda Han, and Weiguang Huo (2025) **Q1 (SJR: 2.540 / JCR Q1 IF: 6.1)**

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Experimental Validation Protocols: Closed-loop sinusoidal and gait-like knee torque tracking benchmarks, alongside Zero-Torque (Transparent Mode) backdrivability and parasitic impedance evaluations.

7. Main Quantitative Results & Statistical Significance

- Closed-Loop Torque Tracking Error: Achieved an RMS torque tracking error (RMSE) of 0.31 Nm.
- Transparent-Mode Parasitic Resistance: Achieved a low resistive backdrive torque of 1.03 Nm during unassisted motion.

8. Internal Study Limitations

1. Physical spring stiffness K_spring is mechanically fixed after fabrication, constraining the upper torque bandwidth.
2. Focuses strictly on low-level electromechanical force tracking without human weighted lifting trials or sEMG evaluation.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Zhao et al. established a benchmark for low-level actuator force tracking (0.31 Nm RMSE), but their controller lacks high-level physiological intelligence to determine optimal assistive torque during fatiguing lifting. Our NMDT architecture couples high-level neuromuscular intelligence (Modules A, B, C) with high-precision low-level actuator tracking.

10. Key Technical Discussions & Methodological Clarifications

Q1: Role of Zhao et al. (2025) as the Low-Level Control Benchmark in Our Review:

High-level neural torque prediction is only effective if the physical actuator accurately tracks the commanded torque without fighting the user. Zhao et al. (2025) establishes our benchmark for closed-loop torque tracking accuracy (0.31 Nm RMSE) and adaptive neural friction compensation (1.03 Nm transparent resistance).

Q2: Trade-Offs Between Series Elastic Actuators (SEA) and Quasi-Direct Drive (QDD) Actuators:

An SEA incorporates a physical elastic element to absorb impact shocks and measure torque via deflection, but the compliant spring limits control bandwidth and adds mass. A low-ratio (9:1–10:1) QDD actuator eliminates the physical spring, achieving higher torque bandwidth and lower backdrive resistance (0.5 Nm in Yu et al., 2025 vs. 1.03 Nm in Zhao et al., 2025).

Q3: Mechanism of the Robust Adaptive Neural Network Force Controller (RANNFC):

Because a slider-crank mechanism has an angle-dependent transmission ratio and non-linear friction, fixed-gain PID control suffers from phase lag and tracking error. RANNFC updates neural weights online to cancel friction and kinematic uncertainties while a sliding-mode term suppresses bounded disturbances.

PAPER #10 (P10)

Q1 (SJR: 0.810) / Q2 (JCR IF: 3.4)

Human-in-the-Loop Optimization of Knee Exoskeleton Assistance for Minimizing User's Metabolic and Muscular Effort

Authors: Sara Monteiro, Joana Figueiredo, Pedro Fonseca, J. Paulo Vilas-Boas, and Cristina P. Santos (University of Minho & University of Porto, Portugal) | **Year:** 2024 | **Journal:** Sensors (MDPI), Vol. 24, Issue 11, Art. 3305 | **DOI:** 10.3390/s24113305

SCImago (SJR): Q1 in Instrumentation & Information Systems (SJR = 0.810) | Clarivate Web of Science (JCR): Q2 (IF = 3.4)

1. Core Idea & Executive Overview

- Problem Statement: Conventional Human-in-the-Loop (HITL) exoskeleton optimization requires wearing an intrusive indirect calorimetry respirometer mask for 20–60 minutes because respiratory gas exchange requires 2–3 minutes per iteration to reach steady state—rendering it impractical for industrial environments.
- Proposed Solution: The SmartOs active knee exoskeleton (modified Technaid H2, 160:1 Harmonic Drive) featuring a real-time Exponential Gaussian Process Regressor (EGPR) that estimates metabolic cost every 10 seconds purely from 4 wearable IMUs (chest, wrist, waist, ankle) + BMI, fused with strain-gauge human–exoskeleton interaction torque (IT) inside a Covariance Matrix Adaptation Evolutionary Strategy (CMA-ES) optimizer.

2. Primary Research Question (RQ)

Can a Human-in-the-Loop (HITL) CMA-ES evolutionary controller minimize both metabolic energy expenditure and muscular effort in real time without a respiratory mask by combining a 10-second IMU-based EGPR metabolic estimator with strain-gauge human–exoskeleton interaction torque?

3. Hardware, Mechanical Design & Actuation Specifications

- Exoskeleton Platform: SmartOs unilateral right-leg active knee exoskeleton (modified Technaid H2, Madrid, Spain; adjustable for heights 1.50–1.95 m and masses 50–100 kg).
- Actuator & Transmission: Maxon EC60 100W Flat Brushless motor coupled to a Harmonic Drive CSD-20-160-2A-GR gearbox (Gear Ratio: 160:1; Nominal Torque: 20 Nm; assists walking speeds 0.5–1.6 km/h).
- Multi-Board Computing Architecture: UDOO X86 Central Controller Unit (Intel Celeron N3160 up to 2.24 GHz running Ubuntu Mate / C++ at 100 Hz) + dual STM32F4-Discovery boards (1 kHz PID control + InertialLab sensor acquisition); 22.4 V 12 Ah LiFePO4 battery.

4. Complete Sensor & Signal Flow Architecture: Sensory Inputs ➔ Computed Outputs ➔ System Utilization

A) Sensory Inputs (Onboard & Lab)

- Active Real-Time Control & HITL Loop: 4× wearable IMUs (InertialLab system at 100 Hz on chest, right wrist, left waist, and right ankle) + User BMI (inputs to EGPR to estimate metabolic cost every 10 s) + Strain Gauges on the exoskeleton frame measuring Human–Exoskeleton Interaction Torque (IT) + Knee Potentiometer (angle) + Motor Hall-Effect Sensors (1 kHz PID torque feedback).
- Offline Ground-Truth Validation Suite (Protocol 1 only): Cosmed K5 portable indirect calorimetry respirometer mask.

B) Computed / Predicted Outputs

- Active Controller Outputs: 10-second metabolic cost estimate $\hat{M}C$ (W/kg), cumulative interaction torque IT, CMA-ES optimal flexion/extension torque peaks ([15, 20] Nm and [-15, -20] Nm), 13-point natural cubic spline torque profile T(P), and 1 kHz closed-loop PID motor command.
- Validation Outputs: EGPR vs. Cosmed K5 RMSE (W/kg), MAPE (%), and Bland–Altman concordance.

C) Output Utilization (Active Control vs. Monitoring)

- ACTIVE REAL-TIME CONTROL: All onboard wearable sensors (4× IMUs + Frame Strain Gauges + Knee Potentiometer + Hall-Effect Torque Sensors) operate actively inside the real-time HITL CMA-ES and PID control loops.
- OFFLINE VALIDATION ONLY: Cosmed K5 respirometer mask (used only in Protocol 1 on 5 subjects without the exoskeleton).

5. Control Architecture & Exact Mathematical Equations

- Real-Time EGPR Pipeline (every 10 s on UDOO X86): 100 Hz 3D acceleration from 4 IMUs → 4th-order Butterworth LP filter (20 Hz) → 10 s windowing → Mean Absolute Deviation (MAD) features → z-score normalization → EGPR (σ = 0.15) → Metabolic Cost (MC).
- Eq. (1) (page 6) — Dual-Objective HITL Cost Function: $CF = \omega_1 \cdot MC + \omega_2 \cdot IT$ (with equal weights $\omega_1 = \omega_2 = 1$)

[Methodological Note: Equation (1) on page 6 contains a typographical repetition 'CF = ω_1 MC + ω_2 MC', whereas the accompanying text explicitly defines the second term as the cumulative human–exoskeleton interaction torque (IT).]

- CMA-ES Optimization (Table 1, page 5): Dimension n = 2 (flexion/extension peaks initialized at ±17.5 Nm); λ = 6 iterations/generation × 20 generations = 120 iterations × 10 s = 20 minutes total optimization time.
- Eq. (2) (page 7) — 13-Point Natural Cubic Spline Trajectory: $T(P) = t_i + a_i(P-p_i) + b_i(P-p_i)^2 + c_i(P-p_i)^3$ tracked by 1 kHz PID (Ziegler–Nichols gains: K_p = 135, K_i = 1.5, K_d = 1.5).

6. Experimental Protocol (Subjects N, Payloads, Repetitions & Conditions)

- Protocol 1 (Real-Time EGPR Metabolic Model Validation): N = 5 healthy volunteers (3M/2F, Age: 25 ± 2.9 yrs, Mass: 77.8 ± 13.4 kg, BMI: 26.8 ± 2.0 kg/m^2) wearing 4 IMUs + Cosmed K5 mask during 10 min standing, 10 min walking (1.5, 2.0, 3.0 km/h), and 10 min sitting.
- Protocol 2 (HITL CMA-ES Exoskeleton Validation): N = 1 healthy female proof-of-concept participant (Age: 23 yrs, Mass: 65 kg, BMI: 24.5 kg/m^2) walking on a treadmill at 1.5 km/h across 3 phases: (1) Acclimatization, (2) 20-min CMA-ES Online Optimization, and (3) Benchmark Comparison (No Exo vs. Zero-Torque vs. Fixed ±20 Nm vs. Optimal HITL).

7. Main Quantitative Results & Statistical Significance

- Protocol 1 (EGPR Metabolic Estimator Accuracy): Offline LOSO cross-validation RMSE = 0.31 W/kg; real-time online validation across 5 subjects achieved RMSE = 0.66 W/kg and MAPE = 26% with a 10-second update window.
- Protocol 2 (HITL CMA-ES Performance): Reduced user metabolic cost by -7.3% (vs. Zero-Torque mode) and -5.9% (vs. No-Exoskeleton), while reducing human–exoskeleton interaction torque (IT) by -32.3%.

8. Internal Study Limitations

1. High-ratio Harmonic Drive (160:1) is non-backdrivable and restricts maximum walking speed to slow gait (0.5–1.6 km/h).
2. CMA-ES requires 20 minutes (120 iterations × 10 s) of continuous steady-state activity to converge for a single task.
3. Full HITL exoskeleton control was validated on N = 1 participant during slow treadmill walking rather than dynamic lifting.

9. Gap of Knowledge & Direct Link to Our NMDT Thesis (Modules A, B, C)

Requiring 20 minutes of evolutionary trial-and-error (CMA-ES) to tune peak torque parameters is impractical for industrial lifting where payload mass and muscular fatigue vary from lift to lift. Our NMDT architecture replaces slow 20-minute black-box evolution with an instantaneous Musculoskeletal Digital Twin (Module A) and continuous single-repetition sEMG fatigue adaptation (Module C).

10. Key Technical Discussions & Methodological Clarifications

Q1: Mask-Free 10-Second Real-Time Metabolic Cost Estimation via EGPR:

Triaxial acceleration signals from 4 wearable IMUs (chest, right wrist, left waist, right ankle at 100 Hz) are low-pass filtered at 20 Hz, segmented into 10 s windows (capturing at least two respiratory cycles), summarized via Mean Absolute Deviation (MAD), z-score normalized, and combined with user BMI inside an Exponential Gaussian Process Regressor (EGPR, $\sigma = 0.15$) running on the UDOO X86 unit.

Q2: Rationale for Combining Metabolic Cost (MC) and Interaction Torque (IT) in Equation (1):

Metabolic cost captures global cardiorespiratory energy expenditure, whereas strain-gauge interaction torque (IT) captures local muscular resistance between the limb and cuffs. Weighting them equally ($\omega_1 = \omega_2 = 1$) minimizes both systemic fatigue and local muscle fighting. (Note: Eq. 1 on page 6 has a print typo 'CF = ω_1 MC + ω_2 MC', whereas lines 310–319 explicitly define the second variable as cumulative interaction torque IT).

Q3: Mechanical Speed Limitation (0.5–1.6 km/h) of the SmartOs Exoskeleton:

The Technaid H2 actuator utilizes a 160:1 Harmonic Drive reducer, which exhibits high reflected inertia and low speed ceiling compared to the 9:1–10:1 Quasi-Direct-Drive (QDD) actuators used in Divekar et al. (2024), Yu et al. (2025), and Xia et al. (2024).